

Lucas Zheng

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EDUCATION

McMaster University

B.A.Sc. in Honours Computer Science Co-op; GPA: 3.9/4.0

Hamilton, Ontario

Sep 2024 – Jun 2029

Relevant coursework: Data Structures and Algorithms; Databases; Concurrent Systems; Computer Architecture; Software Development

EXPERIENCE

Dishon Ltd.

Software Developer Intern

Vaughan, Ontario

May 2026 – Aug 2026

- Developed a replacement for the legacy production data-collection (PDC) system, reducing work-order lookup, page-load, and clock-in/out times by approximately 20× for 200+ machine operators.
- Built a secure API layer for PDC, attendance, and administrative workflows using Next.js, TypeScript, Drizzle ORM, and PostgreSQL; implemented secure session management and role-based access control.
- Implemented audit logging, PostgreSQL full-text search, daily reporting, and support workflows to improve operational traceability and support AS9100- and ISO 9001-aligned recordkeeping.
- Managed production infrastructure for the internal application—Nginx reverse proxies, HTTPS/TLS, internal DNS, PM2, and PostgreSQL migrations—while maintaining 99.85% uptime over two months.
- Within 2 weeks, ramped up on ABAS ERP—including FOP scripting, Screen Editor customization, and REST APIs—to deliver a five-state First Off workflow with traceable work-order status and instructions.

Fortinos Supermarket

Deli Associate

Burlington, Ontario

Sep 2023 – May 2026

- Delivered efficient customer service and collaborated with team members in a fast-paced retail environment.

PROJECTS

TrailSense 🗺️ | Node.js, Flask, TwelveLabs Pegasus/Marengo, Google Maps, OpenAI SDK

- Built a semantic video search engine for mountain-biking footage, enabling riders to retrieve relevant trails from natural-language prompts with timestamped results.
- Integrated TwelveLabs Marengo/Pegasus to generate scene-level embeddings, terrain summaries, and 0–10 difficulty ratings; extracted EXIF GPS/elevation metadata to associate footage with trail locations.

Toronto Airbnb Price Predictor 🏠 | Python, PyTorch, scikit-learn, Pandas, NumPy, Matplotlib

- Trained and evaluated a PyTorch regression model on 15,500+ Toronto Airbnb listings, achieving $R^2 = 0.46$ and \$56 MAE on held-out test data.
- Built a preprocessing and training pipeline with scikit-learn one-hot encoding and feature scaling, Adam optimization, L2 regularization, and validation-based early stopping.
- Visualized results with predicted-vs-actual scatterplots, loss curves, and feature-weight charts.

SyllaBuddy 📖 | Python, TypeScript, Flask, Next.js, Tailwind CSS, OpenAI SDK

- Built a full-stack syllabus analyzer using Next.js, Flask, and OpenAI SDK to extract and summarize course information from uploaded PDFs.
- Developed a Flask API with pdfplumber to parse unstructured syllabi into structured course data, including assessments, deadlines, and textbook information.
- Designed a responsive React and Tailwind CSS dashboard for visualizing parsed course data and automating manual syllabus review and textbook lookup.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, C++, SQL, CSS

Frameworks & Libraries: Next.js, Flask, Tailwind CSS, scikit-learn, Matplotlib, Pandas, PyTorch, NumPy

Databases & Tools: PostgreSQL, Drizzle ORM, Git, Nginx, PM2, ABAS ERP